

iOCO

STOCKPIILING

OPTIMAL STOCKPIILING SOLUTIONS
FOR ESKOM COAL MANAGEMENT



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"Artificial intelligence will permeate every aspect of business in the not-too-distant future, altering fundamentally the nature of business and allowing companies to increase productivity and potentially reduce costs."

JEFF BEZOS

1 Introduction

1.1 Purpose

The purpose of this Business Requirements Document (BRD) is to outline the aims, requirements, and expectations for the Stockpiling Machine Learning Project, commissioned by Eskom. This document serves as a foundation for the project's development and implementation phases, providing a clear understanding of the business needs and technical requirements to ensure the successful delivery of the solution.

1.2 Scope

This Business Requirements Document covers the scope of the Stockpiling Machine Learning Project, including:

- Project Overview: Description of the project, including its aims and expected outcomes.
- Functional Requirements: Detailed specifications of the system's functionalities.
- Non-Functional Requirements: Performance, security, and usability criteria.
- Data Requirements: Information on data sources, types, and formats needed for the project.
- Integration Requirements: Integration points with existing systems and external services.
- Exclusions: Aspects of the project that are not included within this scope, such as specific features or integrations not covered in the initial phase.

1.3 Definitions and Acronyms

- Machine Learning (ML): A branch of artificial intelligence that enables systems to learn from data and make predictions or decisions.
- Stockpiling: The process of accumulating and managing inventory (in this case coal) to ensure consistent supply.
- Horizontal Scaling: Adding more servers or nodes to a system to manage increased load.
- Vertical Scaling: Increasing the capacity of existing servers by adding more resources like CPU, RAM, or storage.

1.4 Background

1.4.1 Problem Overview

Eskom manages multiple power stations with the objective of maintaining optimal coal stock levels. A simplified model is currently used to determine coal stock levels, but it does not adequately account for the complexities of

the supply chain, external factors, individual power station characteristics, and the relationships/dynamics between power stations.

1.4.2 Problem Definition

The primary challenge is to develop a machine learning model that accurately forecasts optimal coal stock levels for each power station, considering a multitude of dynamic and interdependent variables. This model should effectively address the current limitations of the existing approach and provide actionable insights for coal stockpile management.

1.4.3 Desired Outcomes

- Improved accuracy in predicting optimal coal stock levels for each power station, while taking dynamic and interdependent variable into account.
- Enhanced decision-making capabilities for coal stockpile management leading to reduced instances of stockouts and excess inventory. This may lead to potential cost savings from not using the Urgent Procurement Process or overstocking.
- Identification of early warning indicators for potential supply chain disruptions or coal stock level issues.

The Stockpiling Machine Learning Project is initiated in response to Eskom's need for optimal coal stock planning to ensure a reliable power supply. With increasing challenges in inventory management and supply chain disruptions, Eskom seeks a machine learning solution to predict and manage coal stock levels more effectively. The project aims to leverage advanced AI techniques to address these challenges and improve planning and decision-making processes.

2 Business Objectives

2.1 Goals

The primary goals of the Stockpiling Machine Learning project are:

- **Ensure Optimal Coal Stock Management:** Implement a machine learning solution to accurately predict and manage coal stock levels, ensuring efficient inventory control and reducing the risk of stockouts or overstock situations.
- **Enhance Planning Efficiency:** Streamline the process of coal stock management through automated predictions and data-driven insights, leading to improved resource allocation and planning.
- **Improve Decision-Making:** Provide actionable insights and recommendations to support strategic decision-making related to coal stock levels and planning.
- **Ensure Reliability of Power Supply:** Maintain a consistent and reliable supply of coal to meet energy demands, minimizing disruptions and contributing to stable power generation.
- **Stockpile Advisory:** Provide advice on stockpile limits (expected, alarm & minimum) to support and improve planning and decision-making related to coal stock levels.
- **Pattern Recognition:** Recognise and identify patterns, trends and behaviours that affect coal supply and/or burn to enable enhanced stockpile planning and early warning capabilities.

2.2 Business Needs

The Stockpiling Machine Learning Project addresses the following business needs:

- **Accurate Forecasting:** The need of predictions of coal stock level to effectively manage inventory.
- **Data-Driven Insights:** The need to analyse historical data to uncover patterns, trends and anomalies that affect coal stock management.
- **Risk Mitigation:** The need to identify potential risks and anomalies in advance to take pre-emptive measures and avoid/mitigate challenges.

2.3 Expected Benefits

The successful implementation of the Stockpiling Machine Learning Project is expected to deliver the following benefits:

- **Increased Accuracy:** Improved accuracy in predicting coal stock levels, leading to better inventory management and reduced risk of supply chain disruptions.
- **Potential Cost Savings:** Reduced costs from reduced use of the Urgent Procurement Process due to enhanced coal inventory management.
- **Enhanced Operational Efficiency:** Streamlined processes and optimized resource allocation, leading to more effective coal stock management and operational planning.
- **Proactive Risk Management:** Early identification of potential issues and anomalies, enabling proactive measures to mitigate risks and maintain stable coal supply.
- **Strategic Decision-Making:** Data-driven insights and recommendations to support informed decision-making and strategic planning related to coal stock.

3 Requirements

3.1 Functional Requirement

3.1.1 Detailed Descriptions

The functional requirements define the specific behaviours and functions of the Machine Learning (ML) solution for Eskom's coal stock management. The solution should:

- **Predict Coal Supply and Burn:** The model must accurately forecast daily, weekly, and monthly coal supply and consumption rates for each power station.
- **Trend Analysis:** Identify and analyse trends in coal supply and consumption to provide early warning of potential coal stock shortages or surpluses.
- **Scenario Analysis:** Allow users to perform what-if analyses to understand the impact of different variables on coal stock levels.
- **Data Visualization:** Offer intuitive visualizations, such as dashboards and charts, to display current and forecasted stock levels, trends, and analysis results.
- **User Access Control:** Implement user authentication and role-based access control to ensure data security and appropriate access levels.

3.1.2 Use Cases/User Stories

Use Case 1: Trend Analysis

As a data analyst, I need to analyse historical and current trends in coal supply and consumption so that I can provide insights and recommendations for maintaining optimal coal stock levels.

Use Case 2: Scenario Analysis

As an Eskom manager, I need to perform what-if scenarios to forecast the impact of various supply chain disruptions so that I can make informed decisions to mitigate risks.

Use Case 3: Data Visualization

As a user, I need a dashboard to visualize coal stock data so that I can easily monitor stock levels and trends.

3.1.3 User Interface Requirements

- **Dashboard Widgets:** Provide customizable dashboard widgets to display key metrics such as current stock levels, forecasted supply and burn rates, and alerts.
- **Interactive Charts:** Include interactive charts that allow users to drill down into specific data points and view historical trends.
- **Scenario Planning Tool:** Develop an interactive tool within the interface for users to input variables and run scenario analyses, displaying the results in real-time.
- **User Management:** Implement a user management system that allows administrators to create, update, and manage user accounts and roles.

By meeting these functional requirements, the ML solution will provide Eskom with a comprehensive tool to improve forecasting accuracy/enhance predictability and enable proactive decision-making to manage coal stock levels.

3.2 Non-functional Requirements

3.2.1 Performance

- Scalability: The system must efficiently manage large volumes of historical data and data from multiple data sources without significant performance degradation.
- Throughput: The system must process and analyse data at a rate that supports continuous monitoring and forecasting.

3.2.2 Reliability

- Availability: The system must be available, with minimal downtime, to ensure adequate planning and support for Eskom's operations.
- Fault Tolerance: Implement redundancy and failover mechanisms to maintain system functionality in case of hardware or software failures.
- Accuracy: The model should maintain a high level of accuracy in its predictions and analyses, with regular validation and updates to ensure reliability.

3.2.3 Security

- Data Encryption: Ensure that all data, both at rest and in transit, is encrypted using industry-standard encryption protocols to protect sensitive information.
- Access Control: Implement robust access control mechanisms to restrict data access to authorized personnel only, based on their roles and responsibilities.
- Audit Trails: Maintain comprehensive audit logs of all user actions and system events to enable monitoring and investigation of security incidents.

3.2.4 Usability

- User-Friendly Interface: Design the user interface to be intuitive and easy to use, with clear navigation and accessible features for users with varying levels of technical ability.
- Training and Documentation: Provide comprehensive training materials and user documentation to help users understand and use the system effectively.
- Customization: Allow users to customize their dashboards and reports to focus on the metrics and alerts most relevant to their roles.

3.2.5 Scalability

- **Horizontal and Vertical Scaling:** The system architecture should support both horizontal and vertical scaling to accommodate increasing data volumes and user loads.
- **Modular Design:** Implement a modular design that allows for easy addition of new features and integration with other systems as Eskom's needs evolve.
- **Cloud Compatibility:** Ensure the system is compatible with cloud infrastructure to leverage scalability and flexibility in resource allocation.

By addressing these non-functional requirements, the ML solution will provide a robust, secure, and user-friendly system that supports Eskom's coal stock management needs while ensuring high performance and reliability.

3.3 Data Requirements

3.3.1 Data Sources

The system must integrate and process data from various sources to provide accurate and comprehensive analysis. Key data sources include:

- **Coal Supply Data:**
 - Daily, weekly, and monthly volumes of coal supplied to each power station.
 - Identification of mines or suppliers.
 - Mode of transportation (conveyor, rail, truck).
 - Coal quality (proximate analysis, ultimate analysis, calorific value).
 - Contracted coal supply from contracts.
- **Coal Burn Data:**
 - Daily, weekly, and monthly coal consumption at each power station.
 - Forecasted burn volumes based on production schedules.
- **Coal Quality Data:**
 - Proximate analysis data (moisture content, volatile matter, ash content, fixed carbon)
 - Ultimate analysis data (carbon content, hydrogen content, nitrogen content, sulphur content, oxygen content)
 - Calorific value data (gross calorific value, net calorific value)
- **Power Plant Data**
 - Generator Voltage (kV)
 - Turbine Speed (RPM)
 - Steam Temperature (°C)
 - Boiler Pressure (MPa)
 - Heat Rate (kJ/kWh)
 - Coal Consumption (tons)
 - Power Output (MW)
 - Power Station Categorisation (Group 1 or 2)
- **Weather Data:**
 - Daily and historical rainfall data near mines and power stations.
 - Daily temperature readings.
- **Equipment Data:**
 - Planned maintenance activities at mines and power stations.

- Demand Data:
 - Hourly, daily, and monthly electricity demand.
 - Predicted future electricity demand.
- Operational Data:
 - Current and historical coal stock levels at each power station.
 - Expected, minimum, and alarm stock levels for different power station groups.
 - Maximum coal stockyard capacities of all the Power Stations
- Supply Chain Data:
 - Timelines for coal deliveries from suppliers to power stations.
 - Rates at which stock is consumed and replenished.
- External Factors:
 - Dates of major holidays affecting coal supply operations.
- Economic Data:
 - Historical and forecasted coal prices (ZAR/ton).
 - Exchange Rates (ZAR/USD, ZAR/EUR, ZAR/GBP, etc.)

Rationale for Feature Selection

These features have been selected to provide a comprehensive view of the factors influencing coal supply and burn, enabling the model to predict stock levels accurately and manage stockpiles effectively. Each feature contributes to understanding the dynamics of supply and demand, mitigating risks, and ensuring a consistent coal supply to Eskom's power stations. By incorporating diverse data sources, the model can identify patterns, trends, and potential disruptions, facilitating proactive decision-making and efficient stockpile management.

3.3.2 Data Types and Formats

The system must handle various data types and formats, ensuring compatibility and seamless integration:

- Structured Data: Relational databases, CSV files, and spreadsheets containing numerical and categorical data.
- Time-Series Data: Sequential data points collected over time, necessary for forecasting models.
- Unstructured Data: Text logs and records that may require preprocessing and extraction of relevant information.

3.3.3 Data Volume

The system must manage and process large volumes of data efficiently:

- Historical Data: Store and process extensive historical data to support trend analysis and model training.
- Real-Time Data: Continuously ingest and process real-time data to provide up-to-date analysis and alerts.
- Data Storage: Implement scalable storage solutions to handle increasing data volumes over time.

3.4 Integration Requirements

3.4.1 Systems Integration

The ML solution must seamlessly integrate with existing systems at Eskom to ensure efficient data flow and interoperability:

- ERP Systems: Integrate with Eskom's Enterprise Resource Planning (ERP) systems to access relevant data on coal supply, procurement, and inventory management.
- Supply Chain Management Systems: Connect with systems to incorporate delivery data.
- Weather Data Providers: Integrate with external weather data providers to receive up-to-date weather information affecting coal supply and consumption.

3.4.2 External Interfaces

To ensure broad usability and extend functionality, the ML solution must support various external interfaces:

- APIs for Data Ingestion: Develop robust APIs to facilitate automatic data ingestion from disparate systems and data sources, ensuring real-time data flow and minimizing manual data entry.
- APIs for Data Retrieval: Provide APIs for querying model predictions, current stock levels, and risk assessments. These APIs should allow other systems within Eskom's network to integrate and use the ML model outputs for decision-making purposes.
- Custom Connectors: Build custom connectors for systems that do not support standard API integrations, ensuring comprehensive data integration.
- Third-Party Tools: Enable integration with third-party analytical and visualization tools like Power BI, or other business intelligence platforms for enhanced data analysis and reporting.

3.5 Endpoint Consumption and Visualization Requirements

3.5.1 Scenario Analysis and Decision Support Reports:

- Reports
 - Requirement: The system must generate summarising scenario analysis outcomes and recommendations.
 - Details: Reports should include visualizations like charts and graphs for easy interpretation.
- API Endpoints:
 - Requirement: The system must provide APIs that external systems can call to request scenario analysis results.
 - Details: APIs should allow integration with other decision support tools used by Eskom.
- GUI Integration:
 - Requirement: Develop an interactive scenario analysis tool within the user interface.
 - Details: Users should be able to input different variables and assumptions to run what-if scenarios, with results displayed in real-time using visual aids like graphs and tables.

4 Assumptions and Constraints

4.1 Assumptions

Assumptions are conditions that are believed to be true for the purposes of planning and executing the project. They must be validated throughout the project lifecycle to ensure they hold true. The following assumptions have been identified for the Eskom Stockpiling Machine Learning Project:

- **Data Availability:** All necessary data from various sources (e.g., coal supply, burn, weather data, etc.) will be available, accessible, and of sufficient quality for use in the ML model.
- **Stakeholder Support:** It is assumed that all stakeholders, including Eskom management, analysts, and IT support teams, will provide the necessary support and cooperation throughout the project lifecycle.
- **Integration Feasibility:** Integration with existing systems (ERP, SCADA, supply chain management) will be technically feasible without significant issues.
- **Technical Infrastructure:** The existing technical infrastructure such as Azure environments and subscriptions at Eskom is sufficient to support the deployment and operation of the ML solution.
- **Training and Adoption:** End users will be willing to undergo training and adopt the new system for coal stock management.
- **Regulatory Compliance:** It is assumed that the project will comply with all relevant regulatory and legal requirements regarding data security, privacy, and operational standards.

4.2 Constraints

Constraints are limitations or restrictions that may affect the project's progress or outcome. The following constraints have been identified for the Eskom Stockpiling Machine Learning Project:

- **Data Quality and Completeness:** Incomplete, inaccurate, or inconsistent data may hinder the effectiveness of the ML model and its predictions.
- **Budget Limitations:** The project must be completed within the allocated budget, which may limit the scope of activities, resources, and technology choices.
- **Time Constraints:** The project must adhere to the specified timeline, with key deliverables and milestones to be achieved within defined timeframes.
- **Technical Integration:** Integrating the ML solution with existing legacy systems may present technical challenges and require added resources and time.
- **Resource Availability:** Availability of skilled personnel (e.g., data scientists, ML experts, IT support) may be limited, affecting project execution and timelines.
- **Regulatory Requirements:** The project must comply with all relevant regulations and standards, which may impose constraints on data usage, storage, and processing.
- **Change Management:** Resistance to change among stakeholders and users may affect the adoption and effective use of the new system.

5 Acceptance Criteria and Validation

5.1 Criteria

The acceptance criteria define the specific conditions that must be met for the project deliverables to be accepted by the stakeholders. These criteria ensure that the project meets its aims and satisfies the requirements outlined in the BRD. The following acceptance criteria have been established for the Eskom Stockpiling Machine Learning Project:

- **Model Accuracy:** The ML model must achieve a minimum accuracy rate of 95% in predicting coal supply and burn. Accuracy will be validated against historical data and real-time performance.
- **Data Integration:** The system must integrate seamlessly with existing ERP, and supply chain management systems. All relevant data from these systems must be ingested and processed accurately.
- **User Interface:** The user interface must be intuitive and user-friendly, with clear navigation and accessible features. Users must be able to customize their dashboards and reports.
- **Scenario Analysis Tool:** The system must provide a functional scenario analysis tool that allows users to input variables and run what-if scenarios. Results must be displayed in real-time with visual aids such as graphs and tables.
- **Data Security:** All data must be encrypted both at rest and in transit. Access controls must be implemented to ensure that only authorized personnel can access sensitive information.
- **Performance:** The system must manage large volumes of data efficiently. The system must be available, with minimal downtime.
- **Training and Documentation:** Comprehensive training materials and user documentation must be provided. Users must demonstrate proficiency in using the system through training sessions and user acceptance testing (UAT).
- **Post-Deployment Support:** The system must include a post-deployment support plan for ongoing maintenance, updates, and performance monitoring. Support must include regular model validation and retraining, as necessary.

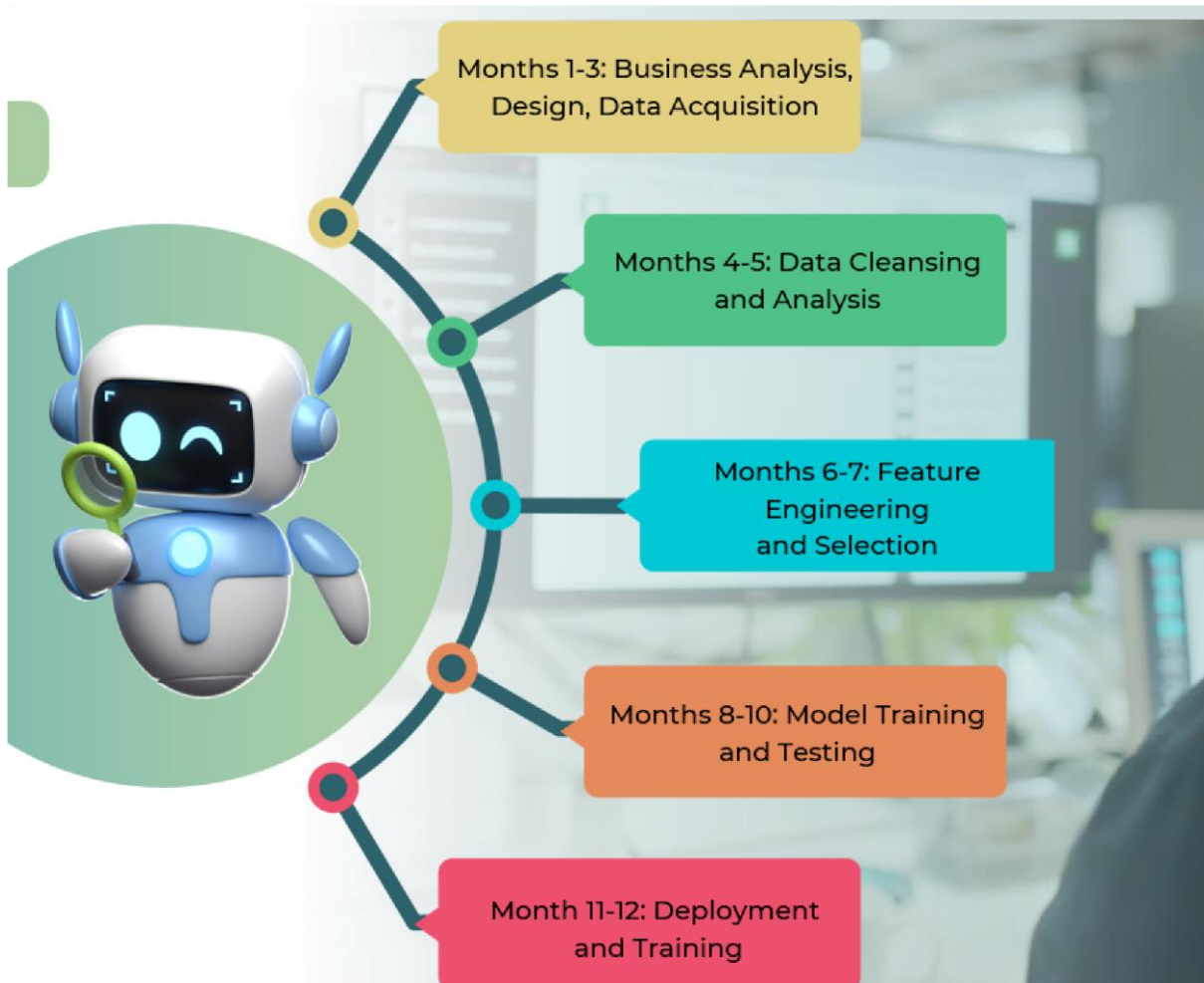
5.2 Validation

The validation process ensures that the project deliverables meet the acceptance criteria and function as intended. The following validation steps will be conducted for the Eskom Stockpiling Machine Learning Project:

- **User Acceptance Testing (UAT):**
 - Conduct UAT sessions with stakeholders, including Eskom management, and data analysts.
 - Collect feedback and make necessary adjustments to the system.
- **Model Validation:**
 - Validate the accuracy of the ML model against historical data.
 - Monitor performance to ensure continued accuracy and reliability.
- **Integration Testing:**
 - Evaluate the integration of the ML solution with existing systems (ERP, supply chain management) to ensure seamless data flow.
 - Address any issues or discrepancies identified during testing.
- **Performance Testing:**
 - Conduct performance tests to ensure the system manages large volumes of data efficiently.

- Measure response times, data processing rates, and system availability.
- Security Testing:
 - Perform security audits to ensure data encryption and access controls are correctly implemented.
 - Evaluate for vulnerabilities and address any security issues.
- Training and Documentation Review:
 - Review training materials and user documentation for completeness and clarity.
 - Ensure that users understand how to use the system effectively.
- Post-Deployment Monitoring:
 - Implement continuous monitoring of the system to track performance and identify any issues.
 - Schedule regular updates and maintenance to keep the system functioning optimally.

6 Implementation Plan



The implementation plan is dependent on several key factors:

- **Data Availability:** Timely access to accurate and complete data from various sources is crucial.
- **Stakeholder Support:** Continuous support and engagement from Eskom stakeholders are essential for successful requirements gathering and system adoption.
- **Technical Infrastructure:** Adequate technical infrastructure must be in place to support data storage, processing, and system deployment.
- **Regulatory Compliance:** Ensuring compliance with data security and privacy regulations throughout the project.
- **User Training and Adoption:** Effective training programs and user engagement are necessary for the successful adoption of the new system.
- **Third-Party Integrations:** Successful integration with external systems and data providers, including weather data services and supply chain management systems.

7 Possible Risks and Barriers

7.1 Data Risks

Data Management Policy Constraints

- Description: Eskom's data management policies may affect the way data is accessed, processed, and used, which in turn could affect the how the Data Science process is implemented or the progress of the project.
- Recommendation: Engagement with the relevant Eskom data management personnel (data stewards, data engineers) is required to develop strategies to ensure that all data processing efforts are compliant with Eskom data policies.

Data Quality Issues

- Description: Incomplete, inaccurate, or inconsistent data may impede the project's viability and success.
- Recommendation: Engaging with Eskom data engineers to ensure that all data sources that will be used in the project are free of quality issues, and to find alternative data sources within Eskom if the identified data quality issues cannot be resolved.

Insufficient Data

- Description: Lack of relevant data may hinder the analysis of data and the development of robust models.
- Recommendation: Simpler models that are less data intensive such as rules-based, optimisation or regression models (or combinations of them) can be used to develop solution that may be viable in some (or all) respects.

Data Integration Challenges

- Description: Combining data from diverse sources may prove to be difficult due to varying data formats/schemas, data refresh schedules, data sourcing, etc.
- Recommendation:

Data Accessibility

- Description: Difficulty in accessing or retrieving data from various sources or systems may be a significant hurdle.
- Recommendation: Resources such as laptops, Azure Data Factory, AWS Glue, VPNs can be provided by Eskom to ensure secure access to their data sources and systems.

7.2 Resource and Infrastructure Risks

Resource constraints

- Description: Limited computational resources (compute clusters, databases & blob storage, etc.) or budget constraints can affect the scope, scale, and delivery of the project. Developing large complex models may require added resources which may in turn lead to increased development and running costs.
- Recommendation: The scope of the deliverable(s) can be limited to ensure that resource constraints are met or alternatively, the relevant stakeholders in Eskom may be approached and requested to provide added resources/funding.

Inadequate Tools and Infrastructure

- Description: Lack of appropriate software or infrastructure can impede the development of the solution (ML model). e.g. using an Azure environment without any model development resources such as Databricks or Machine Learning Studio could result in added difficulty when building ML models.
- Recommendation: Resources such as Databricks, Azure ML Studio, AWS SageMaker, etc. can be provisioned on the chosen cloud stack to make machine learning model development possible.

7.3 Model Development Risks

Model Interpretability

- Description: Complex models (such as neural networks) may be difficult to explain and interpret, making it harder to gain buy-in or trust.
- Recommendation: The following approaches may be followed:
 - Using simpler interpretable models (e.g. linear regression) to develop a solution.
 - Utilising feature importance, SHAP values, etc. to interpret more complex models such as ensemble models (random forests, boosting models, stacking models, etc.), support vector machines, etc.
 - Fitting simpler interpretable models (decision trees, linear regression, etc.) to the outputs of more complex models (neural networks, boosting models, SVMs, random forests, etc.) and using the simpler interpretable model to interpret the outputs of the more complex model.

Model Scalability Issues

- Description: Models or systems that work well with small datasets might struggle when scaled up to larger datasets.
- Recommendation: The following approaches may be followed:
 - Sampling large datasets and training the model on the sampled data. The trained model can then be used to create predictions on the large datasets.
 - Simpler models can be trained on large datasets by using their distributed/hyper-scaled versions, e.g. PySpark, MLlib, Dask, etc.

7.4 Project Management Risks

Change Management

- Description: Implementing new data-driven processes may face resistance from employees or require significant changes in workflows.
- Recommendation: The new data-driven processes can be implemented gradually to allow employees to understand and embrace them, while also allowing them to provide feedback and possible improvements.

Project Scope Creep

- Description: Expanding the project's goals or requirements beyond the initial plan can lead to delays and increased costs.
- Recommendation: Project goals/requirements should be agreed to and documented prior to any development.

Addendum

A Stakeholders

The table below summarises the stakeholders involved in the Stockpiling Machine Learning Project, along with their roles and responsibilities:

Stakeholder	Name	Role	Responsibilities
Project Sponsor	Alwyn van der Merwe	Provides overall direction and funding for the project. Responsible for approving major project milestones and ensuring alignment with organizational goals.	• Approve project charter, budget, and major deliverables. • Provide strategic direction and ensure alignment with organizational goals. • Resolve high-level issues and make critical decisions.
Project Manager	Hlekani Maluleke	Manages the day-to-day operations of the project. Ensures that project objectives are met within scope, time, and budget.	• Develop and manage the project plan. • Coordinate tasks and activities among team members. • Monitor project progress, manage risks, and ensure timely delivery.
Data Engineers	Leza Pela Tshitso Tamane	Develops and implements processes to orchestrate and transform data.	• Perform data orchestration and transformation. • Conduct data cleaning and preprocessing.
Data Analysts	Leza Pela Luthando Mashigo Ntsikelelo Mbekisa	Analyses data to identify trends, patterns, and insights relevant to the project. Ensures data quality and integrity.	• Conduct data cleaning and preprocessing. • Perform exploratory data analysis to inform modelling decisions. • Collaborate with data scientists to refine features and data inputs.
Data Scientists	Callan Abrahams Luthando Mashigo Hloni Motsi	Develops and implements machine learning models. Ensures that models meet business requirements and perform effectively.	• Develop machine learning models according to business requirements. • Train, test, and validate models to ensure performance and accuracy. • Provide insights and recommendations based on model outputs.
Solution Architects	Kashen Verappen Leza Pela Tawanda Chicheko	Manages the technical infrastructure required for the project. Ensures systems are integrated and functioning properly.	• Set up and configure technical infrastructure. • Ensure integration between systems and handle technical issues. • Maintain system security and performance.
End Users	Alwyn van der Merwe Athisha Hardeyal Mateadi Mthimunye	The individuals who will use the machine learning solution in their day-to-day operations. Their	• Test the system and provide feedback on its usability and effectiveness. • Participate in training and onboarding sessions.

		feedback is crucial for system usability.	• Report issues or concerns related to system performance.
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